Paper 4: Overlapping Community Detection: Combining Clique Percolation with Graph Neural Networks

Module 4: Social Network Structure, Measures and Visualization

Type: Method

Research Gaps

- CPM is interpretable but does not scale; GNN methods scale but lose interpretable overlap structure.
- Clique-derived seeding is rarely combined with GNN training objectives.
- Temporal overlapping community detection is under-explored.
- Evaluation practice is inconsistent across metrics, datasets and ground-truth quality.

Key Contributions

- A hybrid CPM-seeded GNN method with soft memberships.
- A temporal extension with a smoothness term for evolving graphs.
- A unified evaluation on real and synthetic overlapping benchmarks.
- Scalability analysis and released code.

Draft Abstract

Clique Percolation (CPM) yields interpretable overlapping communities but scales poorly, whereas graph neural network (GNN) methods scale well but often lose interpretable overlap structure and rarely handle temporal change. This paper proposes a hybrid method that extracts k-cliques as seed communities, initialises node representations from them, and trains a GraphSAGE or GAT encoder with a soft-membership head under a modularity and seed-consistency objective, with a temporal smoothness extension for dynamic graphs. Experiments on SNAP graphs with ground-truth communities (DBLP, Amazon, YouTube), the LFR overlapping benchmark and temporal message networks compare the method with CPM, Leiden, BigCLAM, NOCD and DMoN. The method improves overlapping NMI by [X]% over CPM while reducing runtime by [Y]%, and remains competitive with NOCD in accuracy with better interpretability.

Bracketed values are placeholders to be filled after experiments are complete.

Section-by-Section Outline

1. Introduction

- Overlapping membership in social networks
- Gap: no scalable, interpretable, temporally aware overlap detector
- Research questions and contributions

2. Related Work

- Clique-based and label-propagation methods
- Matrix factorisation (BigCLAM) and GNN-based methods (NOCD, DMoN)
- Dynamic community detection

3. Problem Formulation

- Graph, membership matrix, overlap definition; evaluation criteria

4. Proposed Method

- Clique extraction and seed merging (k = 3 to 5)
- Encoder architecture and soft-membership head
- Loss: modularity plus seed-consistency; temporal smoothness term
- Complexity analysis

5. Experimental Setup

- Datasets: com-DBLP, com-Amazon, com-YouTube, LFR, CollegeMsg, email-Eu-core
- Baselines: CPM, Leiden, BigCLAM, NOCD, DMoN
- Metrics: overlapping NMI, average F1, Omega index, runtime, memory

6. Results

- Accuracy on static graphs
- Scalability with edge count
- Temporal results
- Ablation: seeding, k, loss terms; qualitative case study

7. Discussion

- Interpretability of seeds; sensitivity to overlap level

8. Limitations

- Clique enumeration cost on dense graphs; ground-truth quality

9. Conclusion and Future Work

- Hypergraph and attributed-graph extensions; decentralised platforms

Expected Figures and Diagrams

Keep captions short; place technical detail in axis labels, legends and in-figure labels.

No.	Short caption	Type and content
Fig. 1	Method overview	Pipeline: cliques, seeds, encoder, membership head
Fig. 2	Seed construction	Diagram of k-clique extraction and merging on a toy graph
Fig. 3	Model architecture	Encoder layers and loss components
Fig. 4	Temporal extension	Snapshot sequence with smoothness coupling
Fig. 5	Accuracy comparison	Bar chart of ONMI and F1 across datasets
Fig. 6	Scalability	Runtime and memory versus edge count (log scale)
Fig. 7	Overlap sensitivity	Line plot on LFR with varying overlap level
Fig. 8	Ablation	Bars for seeding, k and loss terms
Fig. 9	Case study	Network visualisation with overlapping nodes highlighted

Related-Work Keyword List

Theme	Search strings
Overlap detection	overlapping community detection; clique percolation; CPM; soft clustering graphs

Theme	Search strings
GNN methods	graph neural network community detection; NOCD; DMoN; GraphSAGE; graph attention network
Factorisation	BigCLAM; nonnegative matrix factorization community detection
Dynamic graphs	dynamic community detection; temporal networks; evolving communities
Evaluation	LFR benchmark; overlapping NMI; Omega index; scalable community detection

References (30) last 5 years

#	Year	Paper	URL
1	2021	Lin, H., Zhan, Y., Zhao, Z., Chen, Y., & Dong, C. “Overlapping Community Detection Based on Attribute Augmented Graph.” *Entropy, 23*(6), 680.	Publisher page orcid
2	2021	“Detect Overlapping Community Based on the Combination of Local Expansion and Label Propagation.” *Algorithms, 14*(8), 237.	Publisher page mdpi
3	2021	El Kouni, I., Karoui, W., & Ben Romdhane, L. “WLNI-LPA: Detecting Overlapping Communities in Attributed Networks based on Label Propagation Process.” *ICSOFT 2021*.	Paper PDF scitepress
4	2021	“Overlapping Community Detection in Temporal Text Networks.” *arXiv preprint arXiv:2101.05137*.	arXiv arxiv
5	2021	“Overlapping community detection using core label propagation algorithm and belonging functions.” *Applied Intelligence*.	Publisher page link.springer
6	2021	“PODCD: Probabilistic overlapping dynamic community detection.” *Expert Systems with Applications, 174*, 114650.	Publisher page sciencedirect
7	2022	He, C., Zheng, Y., Cheng, J., Tang, Y., Chen,	DOI sciencedirect

		G., & Liu, H. “Semi-supervised overlapping community detection in attributed graph with graph convolutional autoencoder.” *Information Sciences, 608*, 1464–1479.	
8	2022	Gupta, S. K., Singh, D. P., & Choudhary, J. “A review of clique-based overlapping community detection algorithms.” *Knowledge and Information Systems, 64*, 2023–2058.	Publisher page link.springer
9	2022	Goswami, S. & Das, A. K. “Determining maximum cliques for community detection in weighted sparse networks.” *Knowledge and Information Systems, 64*, 289–324.	DOI link.springer
10	2022	“Evolutionary multiobjective overlapping community detection based on similarity matrix and node correction.” *Applied Soft Computing, 125*, 109397.	Publisher page sciencedirect
11	2022	“Overlapping community detection using dynamic dilated aggregation in deep residual GCN.” *arXiv preprint arXiv:2210.11174*.	arXiv arxiv
12	2022	“Community detection in overlapping weighted networks.” *arXiv preprint*.	arXiv arxiv
13	2023	“Overlapping community detection on complex networks with Graph Convolutional Networks.” *Computer Communications*.	Publisher page sciencedirect
14	2023	Li, X., Peng, Q., Li, R., & Ma, H. “Dual graph neural network for overlapping community detection.” *The Journal of Supercomputing, 80*, 2196–2222.	DOI link.springer
15	2023	“Community detection based on community perspective and graph convolutional network.” *Expert Systems with Applications*.	Publisher page sciencedirect
16	2023	“Overlapping communities detection through weighted graph community games.” *PLOS ONE, 18*(4), e0283857.	Publisher page pmc.ncbi.nlm.nih
17	2023	Zhao, Y., Deng, K., Liu, X., & Yao, J.	DOI

		“Overlapping Community Detection Algorithm Based on High-Quality Subgraph Extension in Local Core Regions of Network.” *Wireless Communications and Mobile Computing*.	onlinelibrary.wiley
18	2023	“Comparative Analysis of Overlap Community Detection Techniques on Social Media Platform.” *The Computer Journal, 66*(8), 1893–1912.	Publisher page academic.oup
19	2023	“Non-negative matrix factorization for overlapping community detection in directed weighted networks with sparse constraints.” *Chaos, 33*(5), 053111.	DOI pubmed.ncbi.nlm.nih
20	2024	“Overlapping community detection in weighted networks via hierarchical clustering.” *PLOS ONE, 19*(10), e0312596.	Publisher page journals.plos
21	2024	“Comparing the Clique Percolation algorithm to other overlapping community detection algorithms in psychological networks: A Monte Carlo simulation study.”	PubMed record pubmed.ncbi.nlm.nih
22	2024	“A motif-based probabilistic approach for community detection in complex networks.” *Journal of Intelligent Information Systems, 62*, 1285–1303.	DOI link.springer
23	2024	Wu, X., Wang, C.-D., Lin, J.-Q., Xi, W.-D., & Yu, P. S. “Motif-Based Contrastive Learning for Community Detection.” *IEEE Transactions on Neural Networks and Learning Systems, 35*(9), 11706–11719.	DOI pubmed.ncbi.nlm.nih
24	2024	“GNN4DM: A graph neural network-based method to identify overlapping functional disease modules.” *Bioinformatics, 40*(10), btae573.	Publisher page academic.oup
25	2025	Lu, M., Luo, L., & Zhang, X. “Beyond Static Boundaries: Unraveling Temporal Overlapping Communities with Information Bottleneck Guidance.” *ACM Transactions on Knowledge*	DOI research.monash

		Discovery from Data.	
26	2025	“WOCD: A Semi-Supervised Method for Overlapping Community Detection Using Weak Cliques.” *arXiv preprint*.	arXiv arxiv
27	2025	“A Noise-Resilient Semi-Supervised Graph Autoencoder for Overlapping Semantic Community Detection.” *arXiv preprint*.	arXiv arxiv
28	2025	“Hypergraph Overlapping Community Detection for Brain Networks.” *arXiv preprint; accepted for IEEE MLSP 2025*.	arXiv arxiv
29	2026	Bryant, A. G., et al. “Benchmarking overlapping community detection methods for applications in human connectomics.” *Network Neuroscience, 10*(1), 25–61.	PubMed record pubmed.ncbi.nlm.nih
30	2026	“The artificial benchmark for community detection with outliers and overlapping communities (ABCD+o²).” *Journal of Complex Networks, 14*(4), cnag023.	Publisher page academic.oup